ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY

No 2/9, Kamakshipura, Hesaraghatta Hobli, Yelahanka, Bengaluru- 560089
Affiliated to VTU, Belagavi | Approved by UGC/AICTE/Govt. of Karnataka, Bengaluru

ASSIGNMENT – I

Academic Year	2024-25	**Dept.**	Physics
Program	BE	**Date of issue**	15-04-2025
Year / Semester /Sec	2nd	**Date of Submission**	28-04-2025
Course Code	**BPHYS202**		
Course Name	Applied Physics for CSE Stream		

Note	***1. Answer all the questions*** ***2. Each question carries 10 marks.***

Q. No.	Questions	Marks	COs	RBT level
1.	Derive an expression for energy density in terms of Einstein's co-efficient.	10	CO 1	L3
2.	Explain the construction and working of semiconductor laser.	10	CO 1	L2
3.	Explain the application of LASER in laser printer.	10	CO 1	L3
4.	Find the ratio of population of two energy levels in a LASER if the transition between them produces light of wavelength 6493 $\dot{A}$, assuming the ambient temperature at 27 ^{0}C.	10	CO 1	L3
5.	Define Numerical Aperture and arrive at the condition for propagation.	10	CO 1	L2
6.	Obtain an expression for numerical aperture and arrive at the condition for propagation.	10	CO2	L3
7.	Discuss the types of optical fibers based on modes of propagation.	10	CO2	L1
8.	Calculate the numerical aperture and angle of acceptance for an optical fiber having refractive indices 1.563 and 1.498 for core and cladding respectively.	10	CO2	L3
9	State and explain de-Broglie hypothesis. Derive an expression for de-Broglie wavelength by for the matter waves in terms of kinetic energy.	10	CO2	L3
10	State and explain Heisenberg's uncertainty principle. Show that an electron does not exist inside the nucleus on the basis of Heisenberg's Uncertainty Principle.	10	CO2	L2
11	Set up a time independent Schrödinger's wave equation in one dimensional.	10	CO2	L3
12	Calculate the momentum and de Broglie wavelength associated with an electron with a kinetic energy 2.5 keV.	10	CO2	L3

Prepared by:

Prof. Nayan Kumar H N
Assistant Professor
Dept. of Physics

Prof. Pavendhan M
Associate Professor
HOD